

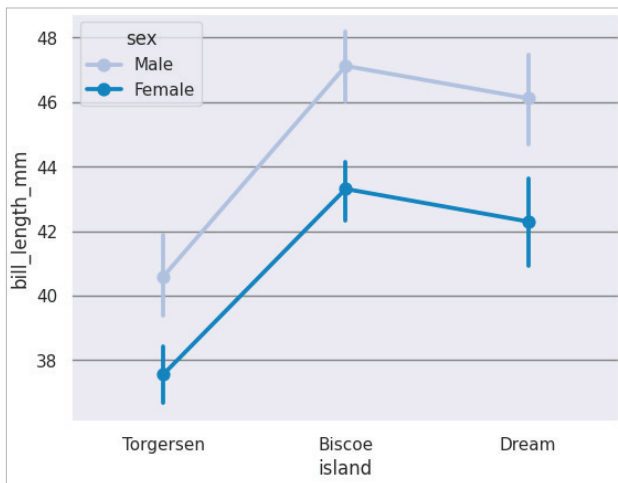


Figure 16: Point plot

```
df,ci=None
)
plt.legend()
plt.show()
```

A histogram is another interesting graph form. It is similar to a bar graph but the intervals are connected and continuous, frequencies are predefined, and the data is not categorical. The code given below can be used to create a histogram using Seaborn (Figure 10).

```
sns.distplot(df['bill_length_mm'],kde = False)
```

Now let's look at a heatmap, which represents the correlation between each feature and another feature or variable, and helps us get a larger picture of the entire dataset. This can be done using the following code (Figure 11).

```
import seaborn as sns
corr=df.corr()
sns.heatmap(corr,annot=True,linewidths=.5,cmap="YlGnBu")
```

The next graph we are going to look at is called a violin plot. It is named so because the graph looks like a violin, and shows the range as well as the frequency of each of the groups. We can get an understanding of the entire group at once. This can be done using the following code (Figure 12).

```
sns.violinplot(x="sex", y="bill_depth_mm", hue="island",
               data=df, palette="muted")
```

A box plot tells us a lot about a feature/variable. We can get to know the range, lowest point, highest point, median and the quartiles using this plot. This can be plotted using the following code (Figure 13):

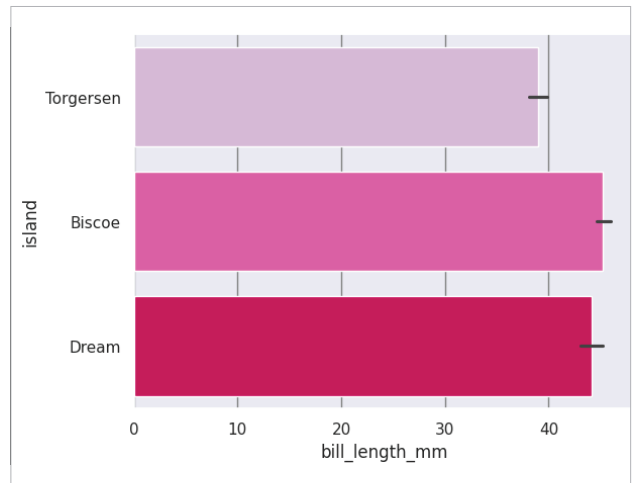


Figure 17: Horizontal bar plot

```
sns.boxplot(x="sex", y="bill_depth_mm", data=df)
```

A regression plot is a scatter plot that also has the regression line. This helps us understand data trends and figure out if a regression algorithm would be helpful to fit the data and train it for an ML model. This can be drawn using the following line of code. Figure 14 shows what the graph looks like.

```
ax = sns.regplot(x="bill_length_mm", y="bill_depth_mm",
                 data=df)
```

A joint plot is used to create a visualisation that combines univariate (histograms or kernel density estimates) and bivariate plots (scatter or regression plots) for two variables in a dataset. This can be plotted using the code shown below:

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.set_style("white")
sns.jointplot(x='bill_depth_mm', y='bill_length_mm',data=df)
```

We can also use the 'hex' kind to create a type of heatmap using hexagons as shown below (Figure 15).

```
sns.jointplot(x='bill_length_mm', y='bill_depth_mm', data=df,
              kind='hex')
```

Pointplot is a function in the Seaborn data visualisation library for Python. It is used to create a point plot, which is similar to a line plot but displays the mean or central tendency of a quantitative variable at different levels of one or more categorical variables. Point plots are useful for showing comparisons between different groups or categories. These can be plotted using the following code (Figure 16).